

**Dr. Thippeswamy L R, NITK, Surathkal**

## **INTERNAL COMBUSTION ENGINES**

**Overview:** This unit consists of 6 lectures. In **lecture 1& 2** and we discuss about the classification of IC engines and the engine parts. In **lecture 3&4** we discuss about the principle of operation of 4-stroke petrol engine and diesel engine. In **lecture 5** we discuss about the principle of operation of 2-stroke petrol and diesel engines. We identify the difference between petrol and Diesel engines. we compare 2-stroke and 4-stroke engines. In **lecture 6** We also discuss about evaluating the performance of engines. In **lectures 7 and 8** we solve some numerical problems on IC engines performance parameters.

**Objective:** At the end of this unit, you should

- Understand the classification of IC engines based on different criteria.
- Understand the functioning of different parts of an IC engine.
- Be able to state the theory behind the working of IC engines.
- Be able to clearly distinguish between 2-stroke and 4-stroke engines, and petrol and diesel engines.
- Be able to evaluate the performance parameters of IC engines and evaluate them.

## **Lecture 1&2**

**Objective:** We discuss about the classification of IC engines and the engine parts.

### **Introduction:**

Any machine, which converts heat energy in to useful mechanical energy, is known as an engine.

The machines may be a gas turbine, steam turbine and an engine.

All the engines comes under two classifications, they are

- i) Internal combustion engine
- ii) External combustion engine

### **Internal combustion engine:**

If the combustion of fuel takes in a cylinder and the heat is converted in to mechanical energy, is known as internal combustion engine, Ex Engines of moped, scooter, bikes, cars, bus, trucks etc;

### **External combustion engine:**

If the combustion of fuel takes place in a combustion chamber and the heat energy is taken to a machine through pipe line there the heat energy is converted in to mechanical energy is known as external combustion engines. Ex, gas turbine and steam engine.

### **Classification of internal combustion engines**

Internal combustion engines are classified according to

1) According to thermodynamic cycle

- i) Otto cycle                      ii) Diesel cycle      iii) Dual combustion cycle

2) According to number of strokes

- i) Two stroke                      ii) Four stroke

3) According to number of cylinders

- i) Single cylinder engine      ii) Multi cylinder engine

4) According to method of ignition

- i) Spark ignition (petrol, gas engine)

- ii) Compression ignition (diesel, vegetable oil)
- 5) According to type of fuel used
  - i) Petrol      ii) Diesel      iii) Gas
  - iv) Bio fuel (peanut oil, sunflower oil, linseed oil)
- 6) According to position of cylinder
  - i) Horizontal engine (hero Honda)
  - ii) Vertical engine (car, bus, truck engines)
  - iii) Vee engine
  - iv) Radial engine (old aeroplane engine)
  - v) Opposed cylinder engine
- 7) According to method of cooling
  - i) Air cooling      ii) Water cooling      iii) Liquid cooling
- 8) According to speed of engine
  - i) Slow speed engine    ii) Medium speed engine    iii) High speed engine

### **Parts of internal combustion engine**

The parts of internal combustion engines are Cylinder, head, piston, piston rings (compression rings and oil control ring), connecting rod and crankshaft.

#### **Function of IC engine parts:**

**Cylinder:** (cylinder block) The cylinder is the main part of an engine. The combustion takes place in the combustion chamber and these gases exert pressure on the piston, due to high gas pressures the piston reciprocates in the cylinder block. The cylinder is designed to withstand high gas pressure. The temperature in the combustion chamber (cylinder block) will reach up to  $2800^{\circ}\text{C}$ . The cylinder has to be cooled properly either by air cooling or water cooling. In case of air cooled engines fins are provided around the cylinder block (Scooter and bikes) in water cooled engines water jackets are provided for the circulation of water to carry away the heat around the cylinder block. The cylinder block material; is aluminium alloy

**Head:** (cylinder head) The head is fitted on the top of the cylinder block. In two stroke engines only spark plug is fitted in the cylinder head.

In four stroke engine inlet valve , exhaust valve and spark is fitted in the cylinder head.- The head is built with two ports, one port, which allows the charge in to the cylinder block, is known as inlet port and the second port that allows the exhaust gases to leave the cylinder block is known as exhaust port. In case of petrol engine spark plug is fitted in the head, in diesel engines fuel injector is fitted to inject the diesel into the cylinder block. The cylinder head material; is aluminium alloy

**Piston:** The piston is a cylindrical plug, which converts heat energy in to mechanical energy. A two stroke piston is fitted with only compression ring. In four-stroke engine both compression ring and oil control rings are fitted. The piston is connecting to the small end of the connecting rod. The piston is made of aluminium alloy. Functions of piston are

- i) The piston will act as a seal
- ii) To provide the passage for heat flow from piston to cylinder block through rings.
- iii) It transmits the force of explosion to the crankshaft through connecting rod.

**Connecting rod:** The small end of the connecting rod is connected to the piston and the big end of the connecting rod is connecting to the crankshaft. The connecting rod converts the reciprocating motion of piston in to rotary motion of crankshaft. The connecting rod is made of I-beam cross section to provide maximum rigidity with minimum weight.

**Crankshaft:** The big end of the connecting rod is connected to the crankshaft. The power transmission starts from the crankshaft. The crankshaft is rigidly fixed in the crankcase. The other end of the crankshaft is connected to a clutch.

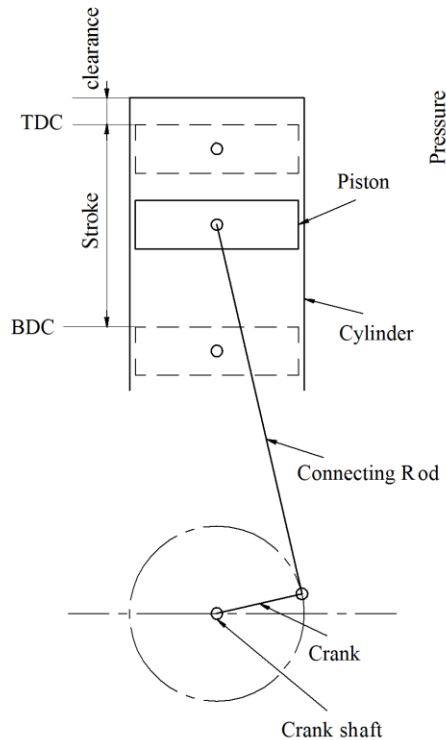
**Crankcase:** Crankcase is fitted at the bottom of the cylinder block. Two-stroke engine crankcase is properly sealed and made airtight. Four stroke engine crankcase will serve as a reservoir, filled with sufficient quantity of lubricating oil. This oil lubricates the main bearings of crankshaft, big end bearings of connecting rod, lubricates the cylinder liner, piston and piston rings.

**Top dead center (TDC):** The top most position of the piston at the cover end is known as top dead center.

**Bottom dead center (BDC):** The lower most position of the piston at the crank end is known as bottom dead center.

**Stroke length (l):** Rotate the crankshaft slowly, the piston starts moving slowly towards the top dead center, Further rotation of crank shaft moves the piston towards the top and suddenly it changes its direction (it starts moving in down ward direction), The momentary stopping of the piston indicates the position of TDC.

Rotate the crankshaft further; the piston starts moving in down ward direction, this movement continues till the piston reaches the bottom dead center. Here also the piston reaches the Bottom and suddenly it changes its direction, the point at which it stops is known as BDC.



**Clearance volume:** When the piston is at TDC position the cylinder volume above it, it is known as clearance volume and it is denoted by  $V_c$

**Swept volume:** The volume swept by piston while traveling from TDC to BDC is known as swept volume and is denoted by  $V_s$ .

$$V_s = (\pi d^2 / 4) l \quad \text{cm}^3$$

**Compression ratio:** It is the ratio of total volume of cylinder ( $V_s + V_c$ ) to the clearance volume ( $V_c$ ). it is denoted 'r'.

$$r = (V_s + V_c) / V_c$$

The compression ratio of petrol engine varies from 7 to 10.

The compression ratio of diesel engine varies from 15 to 24.

**Summary:** We understood the meaning of “IC engines”, we have discussed some of the important classification of IC engines based on different parameters, function of various parts of IC engines, and also looked into some of the important definitions.

**Review questions:**

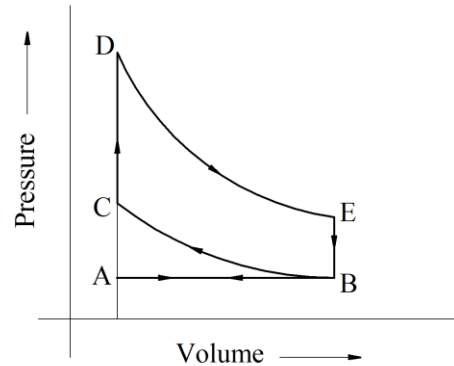
1. What is an internal combustion engine?
2. How do you classify internal combustion engines?
3. Name the important components of internal combustion engine? And state their functions?
4. Define the terms i) Bore ii) Stroke iii) Top dead center iv) Bottom dead center v) Clearance volume, and vi) Compression ratio.

## Lecture 3&4:

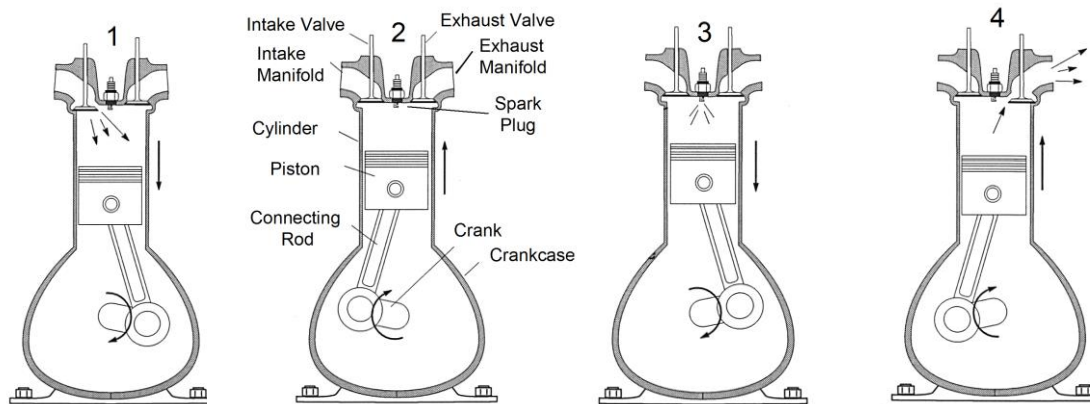
**Objective:** we discuss about the principle of operation of 4-stroke petrol engine and diesel engine.

### Working of four stroke petrol engine

The four-stroke petrol engine works on the principle of Otto (constant volume) cycle. The parts of four-stroke petrol engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, spark plug, inlet and exhaust valve. The four-stroke petrol engine may be air-cooled or water-cooled. The piston performs four strokes to complete one cycle. The four different strokes are i) Suction stroke ii) Compression stroke iii) Power or Expansion stroke iv) Exhaust stroke.



**Suction stroke:** The suction stroke is completed by rotating the crankshaft from  $0^\circ$  to  $180^\circ$ . During suction stroke the inlet valve opens and exhaust valve should be kept in closed condition. When the piston starts moving from TDC to BDC, the volume above the piston increases, resulting in a decrease in pressure (vacuum). This decrease in pressure draws the petrol and air mixture from the carburetor and delivers it to the cylinder; this process is continuous till the pressure inside the cylinder becomes equal to atmosphere. At the end of suction stroke the cylinder is completely filled with petrol and air mixture. At the end of suction stroke the inlet valve closes. The line AB in the PV diagram represents suction stroke (volume of mixture filled in the cylinder).



**Compression stroke:** Rotation of crankshaft from  $180^0$  to  $360^0$  completes the compression stroke. During compression stroke both inlet and exhaust valves are closed. In this stroke the piston travels from BDC to TDC. When the piston starts moving from BDC to TDC the mixture is compressed, and the pressure increases in the cylinder. The line BC represents the compression stroke.

At or near the before the end of the compression stroke, the spark occurs, this spark ignites the petrol and air mix. The combustion of mixture releases hot gases, which will increase pressure at constant volume. The line CD represents increase in the pressure at constant volume.

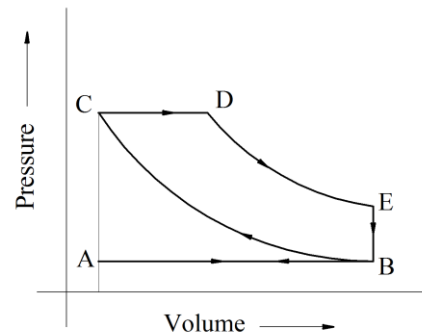
**Power stroke:** Rotation of crankshaft from  $360^0$  to  $540^0$  completes the power stroke. During power stroke (expansion stroke) both inlet valve and exhaust valve are in closed position. The high-pressure gases produced due to combustion, will exert pressure on the top face of the piston, the piston moves rapidly in the down ward direction performs power stroke.

**Exhaust stroke:** Rotation of crankshaft from  $540^0$  to  $720^0$  completes the exhaust stroke. At the beginning of exhaust stroke, the exhaust valve opens, and the upward movement of the piston pushes the exhaust gases out the cylinder. At the end of the exhaust stroke the exhaust valve closes.

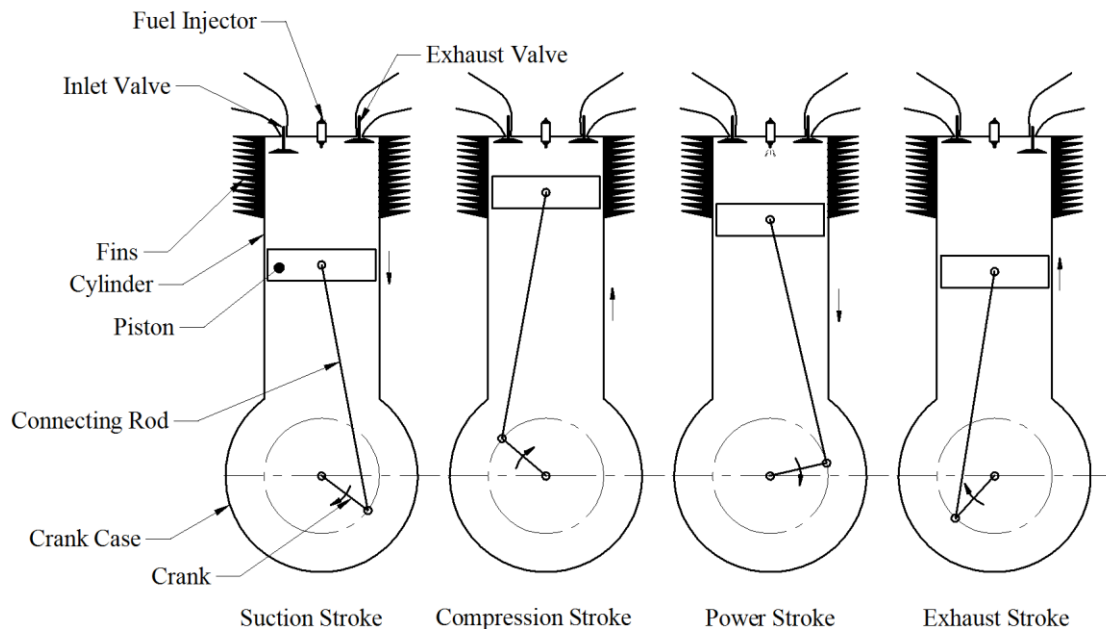
Thus one cycle completes by rotating the crankshaft from  $0^0$  to  $720^0$ .

## Working of four stroke diesel engine

The four-stroke diesel engine works on the principle of diesel (constant pressure) cycle. The parts of four-stroke diesel engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, fuel injector, inlet and exhaust valve. The four-stroke diesel engine may be air-cooled or water-cooled. The piston performs four strokes to complete one cycle.



The four different strokes are i) Suction stroke ii) Compression stroke iii) Power or Expansion stroke iv) Exhaust stroke.



i) **Suction stroke:** The suction stroke is completed by rotating the crankshaft from  $0^\circ$  to  $180^\circ$ . During suction stroke the inlet valve opens and exhaust valve should kept in closed position. When the piston starts moving from TDC to BDC, The volume above the piston increases, results in decrease in pressure (vacuum), This decrease in pressure draws the air from atmosphere and fills the air in to the cylinder, this process is continuous till the pressure inside the cylinder becomes equal to atmosphere. At the end of suction stroke

the cylinder is completely filled with air. At the end of suction stroke the inlet valve closes. The line AB in the PV diagram represents suction stroke (Volume of air filled in the cylinder).

ii) **Compression stroke:** Rotation of crankshaft from  $180^\circ$  to  $360^\circ$  completes the compression stroke. During compression stroke both inlet and exhaust valves are closed. In this stroke the piston travels from BDC to TDC. When the piston starts moving from BDC to TDC the air is compressed, both pressure and temperature of the air increases. At the end of the compression stroke the temperature of the air reaches the ignition temperature of diesel. The line BC represents the compression stroke.

Before the end of the compression stroke, the fuel injector starts injecting the diesel in to the combustion chamber. The heat of the compressed air burns the injected diesel. The combustion takes place at constant pressure. The line CD represents increase in the pressure at constant pressure.

iii) **Power stroke:** Rotation of crankshaft from  $360^\circ$  to  $540^\circ$  completes the power stroke. During power stroke (expansion stroke) both inlet valve and exhaust valve are in closed position. The high-pressure gases produced during combustion, will exert pressure on the top face of the piston, the piston moves rapidly in the down ward direction performs power stroke. The energy is supplied to the flywheel during power stroke. This energy propels the vehicle

iv) **Exhaust stroke:** Rotation of crankshaft from  $540^\circ$  to  $720^\circ$  completes the exhaust stroke. At the beginning of exhaust stroke, the exhaust valve opens, and the upward movement of the piston pushes the exhaust gases out the cylinder. At the end of the exhaust stroke the exhaust valve closes.

Thus one cycle completes by rotating the crankshaft from  $0^\circ$  to  $720^\circ$ , i.e., two revolutions of the crankshaft.

**Summary:** We have discussed about the construction of working of 4-stroke petrol and diesel engines with help of theoretical pressure-volume diagram for the cycles.

### **Review questions:**

1. With a neat sketch explain the working four-stroke petrol engine.
2. With a neat sketch explain the working four-stroke diesel engine.

## Lecture 5 & 6

**Objective:** We discuss about the principle of operation of 2-stroke petrol and diesel engines. We identify the difference between Petrol and diesel engines.

### Two stroke petrol engine.

The parts of two-stroke petrol engine are

**Cylinder:** The cylinder liner is made in the form of barrel (hollow cylinder).

The head is connected at the top of the cylinder. A crankcase is connected at the bottom of the cylinder. The cylinder liner is provided with cooling fins for cooling purpose; an exhaust port is located opposite to the transfer port. One end of the transfer port is connected to the cylinder and the other end is connected to the crankcase through which the charge enters.

**Head:** A spark plug is fitted at the top of the head. Fins are provided in the head for cooling purpose.

**Crankcase:** The crankcase is fitted at the bottom of the cylinder. An inlet port is provided in the crankcase to allow the charge from carburetor to the crankcase, another port is known as transfer port, which transfers the charge from transfer port to the cylinder. The crankcase of two-stroke engine should be an airtight chamber, which prevents the leakage of air in or out of the cylinder.

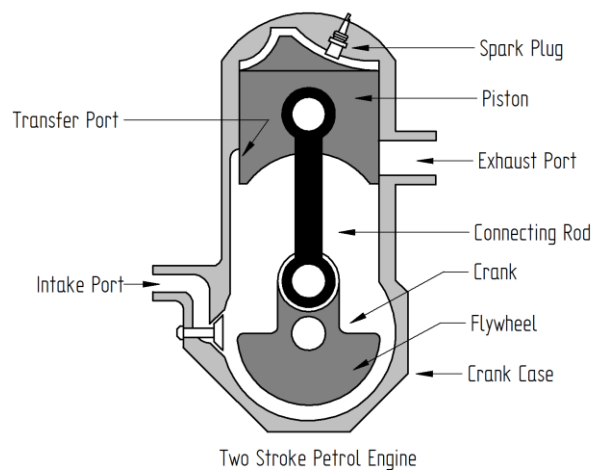
**Piston:** The opening and blocking of the ports are done by the movement of the piston in the cylinder. The piston is fitted with only compression rings (2 or 3 rings)

The two-stroke petrol engine works on the principle of Otto cycle. The parts of two-stroke petrol engine are cylinder, piston, head, crankcase, connecting rod, crankshaft, spark plug, inlet port, transfer port and exhaust port. The piston performs two strokes to complete one cycle. The two strokes are i) First stroke or down ward  
ii) Second stroke or upward stroke.

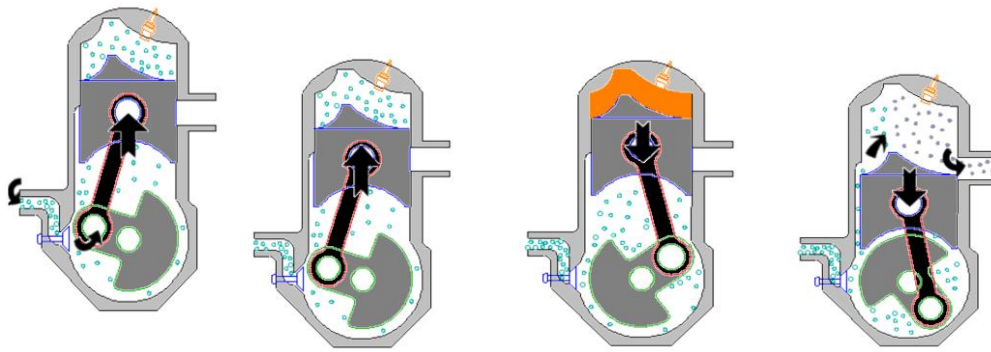
**Upward stroke or working stroke:** In two-stroke petrol engine some charge is present either in the cylinder block or in the crankcase. To start a two-stroke engine, power is supplied either by using a kicker or by electric start.

During upward stroke, the piston reciprocates from top dead center to bottom dead center. As the piston moves upward volume below the piston increases results in decrease in pressure in the crankcase. Due to pressure difference charge (petrol & air) is drawn from the carburetor. As the piston moves further upwards covers both exhaust and transfer port, now the charge is subjected to compression. Before the end of the compression stroke the spark (crank angle  $20^\circ$  before TDC) occurs in the combustion chamber. Due to combustion of charge, the pressure increase, which pushes the piston downwards i.e. the working stroke of the piston. As the piston moves rapidly in the downward direction compress the charge present in the crankcase.

**Down ward stroke or exhaust stroke:** As the piston moves further down wards, first it uncovers the exhaust port. Due to pressure difference the high-pressure gases leaves the combustion chamber. As the piston moves further down wards, it uncovers transfer port, which allows the compressed charge in the crankcase to the cylinder. The fresh charge is deflected upwards by the deflector provided on the top of the piston and pushes the remaining exhaust gases present in the cylinder. The process of removal of exhaust gases from the cylinder is known as scavenging.



First stroke or down ward



## Two stroke diesel engine

### Construction:

**Head-** Only fuel injector is fitted in the head. In air-cooled engine fins are provided in the head for cooling purpose. The head is fitted above the cylinder block.

**Cylinder block-** it consists of 3 ports namely exhaust port, inlet port and transfer port. The piston reciprocates in the cylinder block. The exhaust port is located opposite to transfer port. The piston is fitted with only compression rings. Around the cylinder block fins are provided for cooling purpose.

**Crankcase-** The crankcase is fitted below the cylinder block. The crankshaft is fitted inside the crankcase. The crankshaft is supported by two bearings (ball or roller bearings) on either side of crankshaft.

### Working

**Upward stroke or working stroke:** In two-stroke petrol engine some charge is present either in the cylinder block or in the crankcase. To start a two-stroke engine, power is supplied either by using a kicker or by electric start.

During upward stroke, the piston reciprocates from bottom dead center to top dead center. As the piston moves upward volume below the piston increases results in decrease in pressure in the crankcase. Due to pressure difference air is drawn from the atmosphere. As the piston moves further upwards covers both exhaust and transfer port,

now the air is subjected to compression. The temperature of the air during compression increases just above the ignition temperature of the diesel. Before the end of the compression stroke the fuel is injected and begins to burn. Due to combustion of charge, the pressure increases, which pushes the piston downwards i.e. the working stroke of the piston. As the piston moves rapidly in the down ward direction compress the air present in the crankcase.

**Downward stroke or exhaust stroke:** As the piston moves further down wards, first it uncovers the exhaust port. Due to pressure difference the high-pressure gases leaves the combustion chamber. As the piston moves further down wards, it uncovers transfer port, which allows the compressed air in the crankcase to the cylinder. The fresh air is deflected upwards by the deflector provided on the top of the piston and pushes the remaining exhaust gases present in the cylinder. The process of removal of exhaust gases from the cylinder is known as scavenging. As the piston reaches BDC pushes all the air present in the crankcase, thus completes the down ward stroke.

In the next cycle again the piston moves from TDC to BDC.

### **Difference between petrol and diesel engine.**

- i) The petrol engine works on the principle of Otto cycle and the diesel engine works on the principle of diesel cycle.
- ii) In petrol engine during suction stroke petrol and air mixture is fed in to the cylinder, where as in diesel engine only air is fed in to the cylinder.
- iii) In petrol engine, the carburetor prepares correct air fuel mix and is delivered to the cylinder, In case of diesel engine fuel injector injects diesel directly in to the cylinder.
- iv) In petrol engine the air fuel mix is ignited by the spark given by the spark plug. In case of diesel engine, the heat of the compressed air burns the injected diesel in cylinder.
- v) The compression ratio of petrol engine varies from 6 to 9, whereas in case of diesel engine the compression ratio varies from 15 to 24.
- vi) The petrol engine is cheaper than a diesel engine.
- vii) The maintenance cost of diesel engine is more than a petrol engine.

- viii) The petrol engine gives more pollutants than a diesel engine.
- ix) The petrol engine runs smoother than a diesel engine.
- x) The petrol engine produces less vibration than a diesel engine.
- xi) The diesel engine gives more mileage than a petrol engine.

**Summary:** we have discussed about the construction and working of two stroke petrol and diesel engines with the help of sketches. We have also discussed about the differences between petrol and diesel engines.

**Review questions:**

1. With a neat sketch explain the working two-stroke petrol engine?
2. With a neat sketch explain the working two-stroke diesel engine?
3. What are the differences between petrol engines and diesel engines?

**Difference between two stroke and four stroke engines.**

- i) In 4 - stroke engine, one cycle is completed in two complete revolution of crankshaft ( $720^\circ$ ) where as, in a 2 - stroke engine one cycle is completed in one complete revolution of the crankshaft ( $360^\circ$ ).
- ii) In 4 - stroke engine, the suction and exhaust ports are opened by valves, whereas in 2 - stroke engine the piston itself opens and blocks the ports during reciprocation.
- iii) The crankcase of a 4 - stroke engine is a closed chamber, where as in 2 - stroke engine the crankcase is closed pressure tight chamber.
- iv) In 4 - stroke engine the turning moment is not uniform, so a heavy flywheel is required. Whereas in 2 - stroke turning moment is uniform, so a lighter flywheel is required.
- v) In 4 - stroke engine During exhaust stroke the movement of the piston pushes all the gases out of cylinder, whereas in 2 - stroke engine removal of exhaust gases is partly due to pressure difference and partly due to scavenging.
- vi) In 4 - stroke engine wear & tear is less because of one power stroke in two revolutions of crankshaft, whereas in 2 - stroke engine wear & tear is more because of one power stroke in one revolution of crankshaft.
- vi) In 4 - stroke engine the volumetric efficiency is better, whereas in 2 - stroke engine the volumetric efficiency is poor.
- vii) In 4 - stroke engine the thermal efficiency is better, whereas in 2 - stroke engine the thermal efficiency poor.
- viii) In 4 - stroke engine due to one power stroke in two revolutions lesser cooling is needed, whereas in 2 - stroke engine due to one power stroke one revolution of crankshaft more cooling is needed.
- ix) In 4 - stroke engine the charge directly enters into the cylinder, whereas in 2 - stroke engine first the charge enters into crankcase then through transfer port it enters in to the cylinder.
- x) In 4 - stroke engine the oil consumption is less, whereas In 2 - stroke engine oil consumption is more.
- xi) In 4 - stroke engine number of components are more hence it is costly, whereas in

2 - stroke engine number of components are less hence it is cheap.

xii) The 4 - stroke engine needs more maintenance than a 2 – stroke engine.

xiii) The 4 – stroke engine needs more space than a 2 – stroke engine.

### **Simple calculations in internal combustion engines.**

i) **Mean effective pressure ( $p_m$ ):** It is defined as the average pressure is acting on the piston during the entire expansion (power stroke) stroke.

$$p_m = \text{Mean effective pressure } \text{N} / \text{m}^2$$

ii) **Indicated power (IP):** The power developed within the piston –cylinder arrangement by the combustion of fuel is known as the indicated power. The pressure acting on the piston varies throughout the working cycle. To record the variation of pressure for one cycle of operation, a device called *piston indicator* is mounted by drilling a small hole on the cylinder cover. It mainly consists of a small plunger and a cylinder. The plunger displacement is proportional to the pressure acting on it from inside against the spring force on the other side. The movement of the plunger transmitted to a stylus through linkages. The stylus traces out a graph on a recording drum, which rotates at a constant speed. The graph thus obtained is called the indicator diagram. The area of the indicator diagram is proportional to the work done in a cycle.

Mean effective pressure ( $p_m$ ):

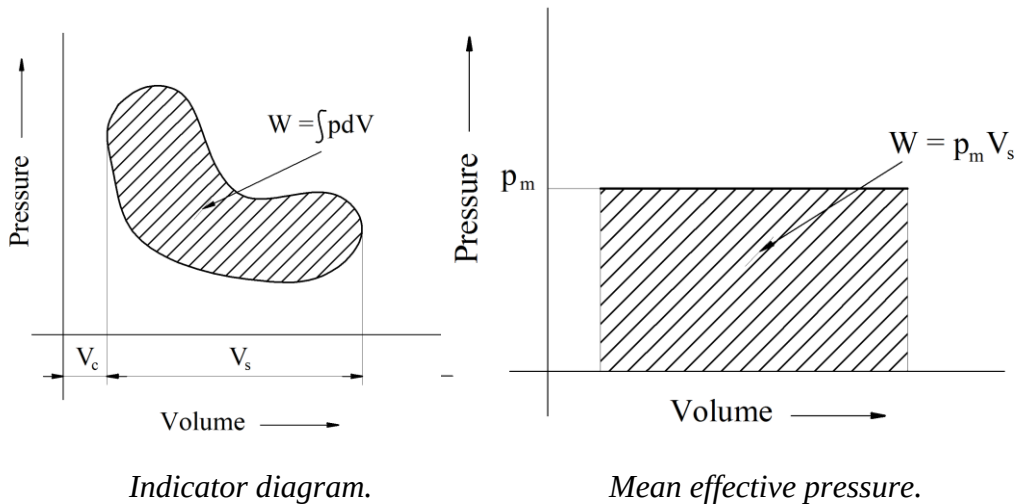
The work done on the piston for one cycle of operation is given by

$$W = \int p dV,$$

Where the integration is carried out for one cycle,

$p$  is the pressure,

$V$  is the volume.



The right hand side of the equation is nothing but the area within the loop on the pressure-volume diagram.

The mean effective pressure is defined as the equivalent constant pressure which has to be acting on the piston during the expansion stroke, to give the same work output as the varying pressure, in one cycle.

From the indicator diagram, the mean effective pressure can be calculated as,

$$p_m = s \cdot a / l.$$

where,

$s$  = spring constant of the spring used in the piston indicator,

$l$  = length of the indicator diagram,

$a$  = area of the indicator diagram.

*Note that, spring constant is the pressure required to cause unit deflection of the spring.*

i) When  $p_m$  is expressed in **N/ m<sup>2</sup>**

$$IP = \frac{p_m L A n}{60 \times 1000} \quad \mathbf{kW}$$

ii) When  $p_m$  is expressed in **bar**

$$IP = \frac{100 p_m L A n}{60} \quad \mathbf{kW}$$

Where,

$p_m$  = mean effective pressure,

$L$  = stroke length,

$A$  = area of cross-section of the piston,

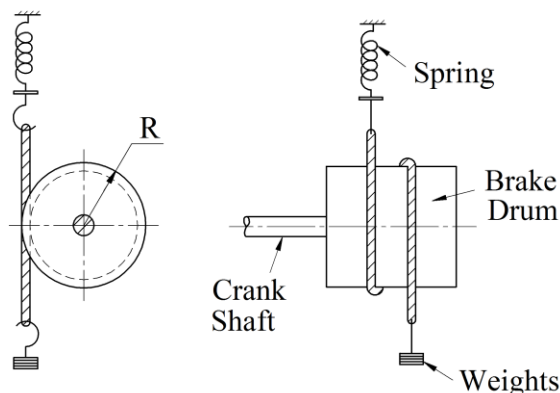
$n$  = number of cycles per minute,

=  $N/2$  for a four stroke engine,

=  $N$  for a two stroke engine.

$N$  = crank shaft speed, rpm

ii) **Brake power (BP):** The power available at the crank shaft is always less than the power developed within the piston-cylinder arrangement because of frictional losses in the moving parts. The power actually available at the crank shaft is called the brake power. It can be measured using *dynamometers*. One such dynamometer is the *brake-drum* dynamometer.



It consists of a drum, which is mounted on the crankshaft. A rope is wound on the drum. One end of the rope is connected to a spring balance, and the other end, to a weight-loading device.

The torque on the brake drum is given by,

$$T = (W - S) \times R$$

Where,

$W$  = weight on the rope, N.

$S$  = spring balance reading, N.

R = mean radius of brake drum, m.

Brake power is given by,

$$BP = \frac{2 \pi N T}{60 \times 1000} \quad \mathbf{kW}$$

iii) **Frictional power:** The difference between indicated power and brake power is known as frictional power

$$FP = IP - BP \quad \mathbf{kW}$$

iv) **Mechanical efficiency:** It is defined as the ratio of brake power to indicated power

$$\eta_{\text{mech}} = \frac{\text{Brake power}}{\text{Indicated power}}$$

v) **Thermal efficiency:** In IC engines, energy is supplied to the engine by burning fuel. But all of the energy that is supplied is not converted into useful mechanical work. Some of the energy supplied is lost through hot exhaust gases, some due to the cooling of the engine and some through radiation and convection heat losses. The fraction of the energy supplied that is available as useful work determines the thermal efficiency of the engine. The thermal efficiency can be calculated either for the indicated power, or for the brake power. Accordingly they are referred to as indicated thermal efficiency, and brake thermal efficiency.

Heat supplied to the engine per sec = mass of fuel burnt x calorific value.

$$\eta_{\text{indicated thermal}} = \frac{\text{Indicated power}}{m_f \times CV}$$

$$\eta_{\text{brake thermal}} = \frac{\text{Brake power}}{m_f \times \text{CV}}$$

Where,

$m_f$  = Mass of fuel used in **kg / sec.**

CV = Calorific value of fuel **kJ / kg**

vi) Specific fuel consumption (SFC):

It is the mass of fuel supplied per hour in order to get unit power output.

$$\text{SFC} = \frac{m_f}{\text{Power}} \text{ kg/kW-hr}$$

SFC can be calculated on indicated power basis or on brake power basis.

**Summary:** We have discussed the differences between 2-stroke and 4-stroke engines.

We studied how various performance parameters of IC engines are evaluated.

**Review questions:**

1. Define i) Mean effective pressure ii) Indicated power iii) Brake power  
iv) Mechanical efficiency v) Indicated thermal efficiency vi) Brake thermal efficiency
2. Explain how indicated power and brake power are measured in an IC engine.